



AI for Strategy & Research Summaries

Week 2 • Day 4 • Strategic Design & Metrics

TPM Academy



Day 4 Objectives

- Use AI to **synthesize multi-source research** with traceable evidence
- Detect and correct the four **summarization failure modes**: theme inflation, citation invention, premature confidence, source bias
- Stay on the **safe side of the bright line**: structure source material, never ask AI to invent market facts
- Produce a **provenance log** that makes your AI use auditable
- Assemble an **AI-Assembled Strategy Brief** for your quad's problem



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Re-ground in your Day-3 audit + design principles
09:15 – 10:30	Block 1 + Activity 1	Multi-source synthesis drill
10:45 – 12:00	Block 2 + Activity 2	The Research Pack pass
13:00 – 14:15	Block 3 + Activity 3	Competitive snapshot
14:30 – 16:00	Block 4 + Activity 4 + Readouts	Brief assembly + provenance

Block 1 — Strategy-Scale Summarization

Beyond a single document, beyond a single prompt



What's Different at Strategy Scale

Week 1 prompting	Strategy-scale prompting
Single document or short brief	5–15 documents, often contradictory
One prompt, one output	Iterated prompts, validated output
"Summarize this"	"Synthesize across these, tag evidence"
You verify confidently in 5 minutes	You only catch fabrication if you're looking

Mindset shift: at strategy scale, you are not consuming the AI's output — you are *auditing* it. The unit of work is the prompt + the validation pass.



Four Failure Modes (the ones that bite)

Failure	What it looks like	Fix
Theme inflation	3 distinct complaints lumped under "usability"	Demand verbs, not categories; cap themes at 5
Citation invention	"Ticket T-16" doesn't exist (pack stops at T-14)	Verify every cited source exists; demote uncited claims
Premature confidence	Plausible synthesis with no acknowledgment of gaps	Force a "what's missing" prompt as second turn
Source bias	The AI weights the loudest source highest, regardless of accuracy	Ask: "How would your synthesis change if I removed Source X?"



The Validated Synthesis Prompt

Role: Research analyst supporting a TPM.

Context: Below are [N] interview transcripts and [M] support tickets from [user segment]. I am investigating [problem statement].

Task: Extract the top 5 recurring pain themes.

Constraints:

- Each theme must cite 2+ specific source identifiers (e.g., "Interview A3, Ticket T-04")
- Themes must be verbs, not categories
- Flag any theme you inferred without direct evidence
- At the bottom, list 3 questions the sources do NOT answer

Format: Markdown list; each entry: Theme / Evidence / Confidence (H/M/L) /

Inferred? (Y/N). End with "What's missing" section.

Activity 1 — Multi-source Synthesis Drill

Quads • 35 minutes

Warm-up: a small healthcare-scheduling pack (different domain). Single-prompt attempt, then iterate after observing failure modes.

- 10 min: single-prompt attempt + run
- 10 min: inspect — which of the 4 failure modes appeared?
- 15 min: iterate the prompt; re-run; compare



10 attempt • 10 inspect • 15 iterate

Block 2 — The FieldPulse Research Pack

Real materials, real noise, real strategy implications



What's in the Pack

- 3 anonymized dispatcher interview transcripts (~5 pages each)
- 14 support tickets (already familiar from Week 1 Day 4)
- 1 industry analyst report (5 pages)
- 2 competitor product tour transcripts
- 1 internal "voice of customer" Slack export (33 messages)

Be honest: the pack is noisy, partly contradictory, and has gaps. That's the point. Real strategy work feels like this.



The Cross-Validation Step

Each quad member takes a different theme produced by the AI and verifies **every** citation against the source. Mark any that fail.

- If a citation can't be verified → **cut the citation**, lower theme confidence to L
- If 2+ citations cut from one theme → **cut the theme** or relabel as "hypothesis to validate"
- If a quote is paraphrased misleadingly → **flag** in provenance log

Don't trust: the AI's confidence rating about its own output. Independent verification is the only signal.

Activity 2 — The Research Pack Pass

Quads • 40 minutes

Use your iterated prompt on the real FieldPulse Research Pack. Cross-validate every citation.

- 5 min pre-flight (confirm prompt has constraints)
- 15 min run (split pack across 2–3 calls if needed)
- 15 min cross-validate (one theme per member)
- 5 min edit (cut failed themes; lower confidence)



5 prep • 15 run • 15 verify • 5 edit

Block 3 — The Competitive Snapshot

The brightest line in AI-assisted research

The Bright Line

**AI can summarize and structure source material you provide.
AI cannot tell you what your competitors actually do.**

Safe (you provide source)

- "Structure these four transcripts into a matrix"
- "What 3 questions do these not answer?"
- "What's missing vs. our product?"

Unsafe (AI invents)

- "What is ServiceTitan's market share?"
- "Who is the leader in dispatch SaaS?"
- "Cite a study on dispatcher productivity"



The Two-Prompt Pattern

Prompt 1 — Structure what we know:

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Role: Competitive analyst.  
Context: Pasting in 2 competitor product tour transcripts plus  
their public pricing pages.  
Task: Build a comparison matrix.  
Constraints:  
- Use only facts from the materials I pasted  
- Do NOT pull from your training data  
- Flag any cell where the source is silent  
Format: Markdown table, with citation column.
```

Prompt 2 — Tell us what's missing:

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From the matrix above, what 3 questions about these competitors  
do the source materials NOT answer? For each, suggest where a  
TPM should look (analyst reports, customer interviews, public  
earnings calls).
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Activity 3 — Competitive Snapshot

Quads • 40 minutes

Use the two-prompt pattern on the competitor materials in the Research Pack. Build a matrix and a list of unanswered questions.

- 15 min: structure-what-we-know prompt + verify cells
- 15 min: what's-missing prompt + decide which gaps matter
- 10 min: write the Competitive Snapshot section of your brief



15 structure • 15 missing • 10 write

Block 4 — Brief Assembly + Provenance

Stitch into a brief, write the parts AI cannot, log the parts AI did

The Strategy Brief Structure

1. **Executive summary** — hand-written, 5 bullets max
2. **Top 5 user pain themes** — AI-summarized, evidence-tagged, cross-validated
3. **Competitive snapshot** — AI-drafted from your materials, hand-edited
4. **Strategy implications** — hand-written; ties themes to NS and design principles
5. **Provenance log** — mandatory; documents every AI use and validation step

Hand-written sections matter most: they are where the TPM's judgment shows. AI does the volume; you do the meaning.



The Provenance Log

A trust-building artifact: documents *where* AI was used, *which* prompts, *what* validation was done, and *where* the AI was wrong.

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## Provenance log

### AI-generated sections
| Section | Prompts (link) | Validation | Findings |
|-----|-----|-----|-----|
| Pain themes | Pattern Library #3 | All citations verified | Cut 1 hallucinated theme |
| Competitive matrix | Pattern Library #5 | Manual check vs source | Flagged 2 silent cells |
| Missing questions | Pattern Library #6 | Quad consensus | Adopted 2 of 3 |

### Where AI was wrong
- Lumped "reconcile" and "timecard" into single theme; we split them
- Inferred a "training" pain theme with no source; cut entirely
- Cited "Ticket T-16" — does not exist; replaced with T-11

### Manual hand-written sections
- Executive summary
- Strategy implications
- This log
```

Activity 4 — Brief Assembly + Provenance Log

Quads • 45 minutes • Readouts

Stitch the day's outputs into a single brief. Hand-write the executive summary and strategy implications. Complete the provenance log.

- 15 min: hand-write executive summary
- 10 min: hand-write strategy implications
- 10 min: complete provenance log
- 10 min: polish for sharing



15 exec • 10 strategy • 10 log • 10 polish • 90s readouts



Day 4 Wrap

What we built

- Validated synthesis prompt (Pattern Library entry)
- Two-prompt competitive pattern
- Cross-validation discipline
- AI-Assembled Strategy Brief with provenance

What opens tomorrow

- **Mapping customer journeys to features**
- Friday readouts: full Week-2 strategy package
- Bridge to Week 3 Product Requirements Documents (PRDs)

Bring to Day 5: Your Strategy Brief, NS, and 3 design principles. All three feed the journey map directly.



Questions & Reflection

Where in your brief did the AI's confidence exceed your evidence?